

Tomorrow's Newspaper

The making of EON, a machine that reads annual reports, and an honest test of what it could see



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Abstract

The pre-specified post-cutoff test did not establish a BUY-minus-SELL return spread: the six-month difference was 4.6 percentage points ($p = 0.094$). A later, post-hoc rank test found a small association, but it does not replace the failed primary test. An options scan also showed an association with subsequent absolute excess returns; without a volatility baseline, that remains an exploratory finding rather than demonstrated skill.

These results came from seventeen months of building EON, a system that reads annual reports and records structured judgements. Five versions produced 29,375 stored answers, including 6,973 three-lens readings in the main EON database. The engineering problem was to make scarce model requests useful: acquire readable filings, coordinate parallel workers, validate answers, and retain progress through interruptions.

The paper follows that construction, evaluates the resulting archive, and examines the model's failures. Historical performance weakened when the tests separated what the model might already know from what remained uncertain. The durable contribution is a working research system and an auditable account of its limits. A frozen ledger of 1,257 verdicts defines a prospective test for October 2027.

Research, not investment advice.

1. A reader for every annual report

Every public company in the United States files a 10-K each year. It is the fullest account a company gives of itself: what it sells, how it makes money, what could go wrong, what it owes. The median 10-K in this project runs to 141 pages. Thousands are filed every spring. Professional analysts read the few they are paid to cover; almost nobody reads the rest.

I wanted a reader that never tires. Give it any filing and it would read the whole thing, ask the same careful questions every time, and write down what it concluded, so that I could come back a year later and see whether it had been right. A language model can now read 141 pages in a minute. The hard part is everything around that minute.

One filing becomes one countable verdict

Specimen: Apple 10-K for fiscal 2025, filed 31 Oct 2025 / first stored reading

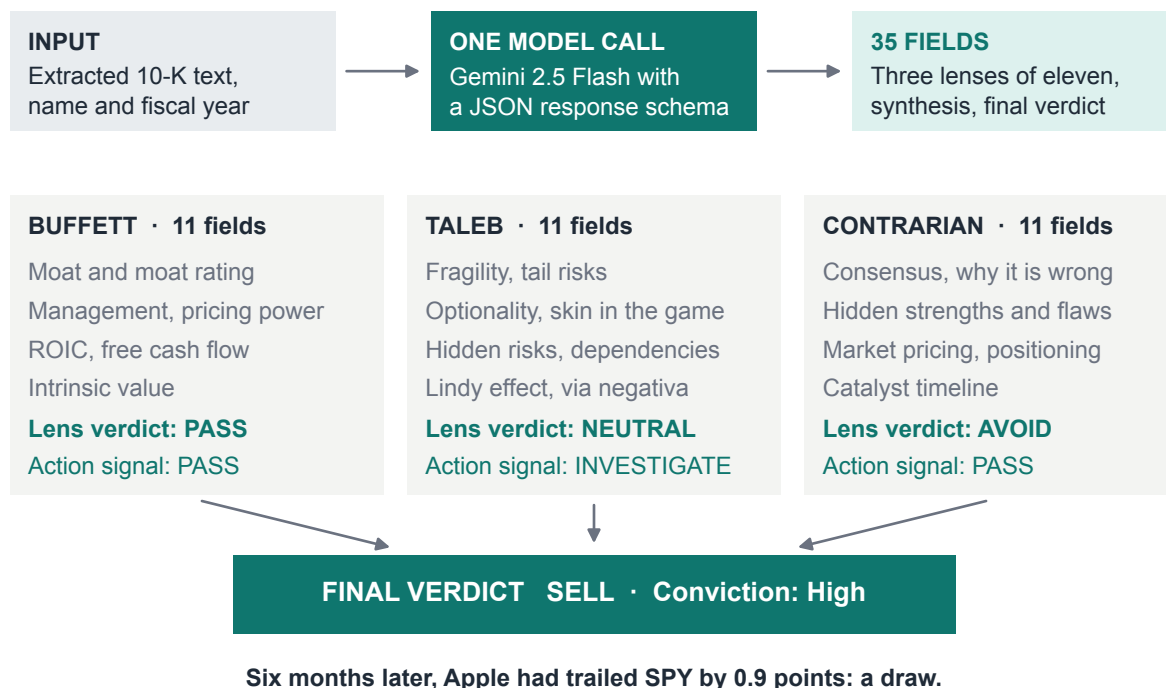


Figure 1: EON's first stored reading: Apple's 10-K for fiscal 2025, filed 31 October 2025. The model reads the filing as Warren Buffett, Nassim Taleb and a contrarian would, then gives one verdict. Six months later Apple had trailed the S&P 500 by 0.9 points: a draw.

Figure 1 is what the finished machine produces. Its verdict on Apple reads: "SELL. Conviction: High. Apple is an undeniably excellent company, but a high conviction SELL rating is warranted given the significant disconnect between its robust fundamentals and its highly stretched valuation." It is specific, fluent and confident. Whether it is right is a separate question, and Part II is about how hard that question turned out to be.

The paper has three parts. Part I tells how the machine was built, version by version, and why each component exists. Part II describes everything it was asked and tests its answers against what happened. Part III is about what the project taught: where the model failed, and why the whole thing was worth doing.

Part I · Building the reader



Illustration — The workshop. A metaphor for the system in Figure 3: the machinery matters because the reader cannot run alone.

EON was built five times. Each version answered a failure of the last and kept what worked (Figure 2).

2. The workshop: learning from great companies

The project did not begin as a stock picker. It began, in May 2025, as a question about excellence: what do great companies have in common, and which other companies have it too?

The first answer was a folder of Python scripts called `10K_automator`. They downloaded up to thirty years of annual reports for 39 long-run compounders (Apple, Costco, Copart, Fair Isaac, O'Reilly, AutoZone and others), converted each one to PDF with a headless Chrome browser, extracted the text, and asked Gemini 2.5 Flash, then an April preview, to describe what made the company work that year. A second pass distilled each company's thirty readings into success factors; a third distilled all 39 into a meta-analysis.

It found six “universal” factors: operational excellence, capital allocation, continuous innovation, domain expertise, culture, and customer focus. The model credited continuous innovation to 100% of the companies. That should have been a warning. A factor that every great company has, and that nearly every annual report claims, cannot tell great companies from anyone else.

The scripts then read the recent filings of about 1,900 other companies and scored each from 0 to 100 on its resemblance to the pattern. Between 3 and 13 May they produced 11,741 readings of individual filings, 4,641 of them on 13 May alone. Two more experiments followed: a contrarian scan on 26 May, scoring 1,933 companies on six dimensions of hidden strength, and an options scan on 4 June that proposed 951 trades in calls or puts, run on ten API keys with ten progress files so that a crash would not lose the day.

Almost everything in EON started in those five weeks: reading whole filings through a browser, parallel workers, progress files that survive a crash, and the ambition to read the entire market.

Five versions, one idea carried forward

May 2025 to June 2026 / each version kept what the last one taught

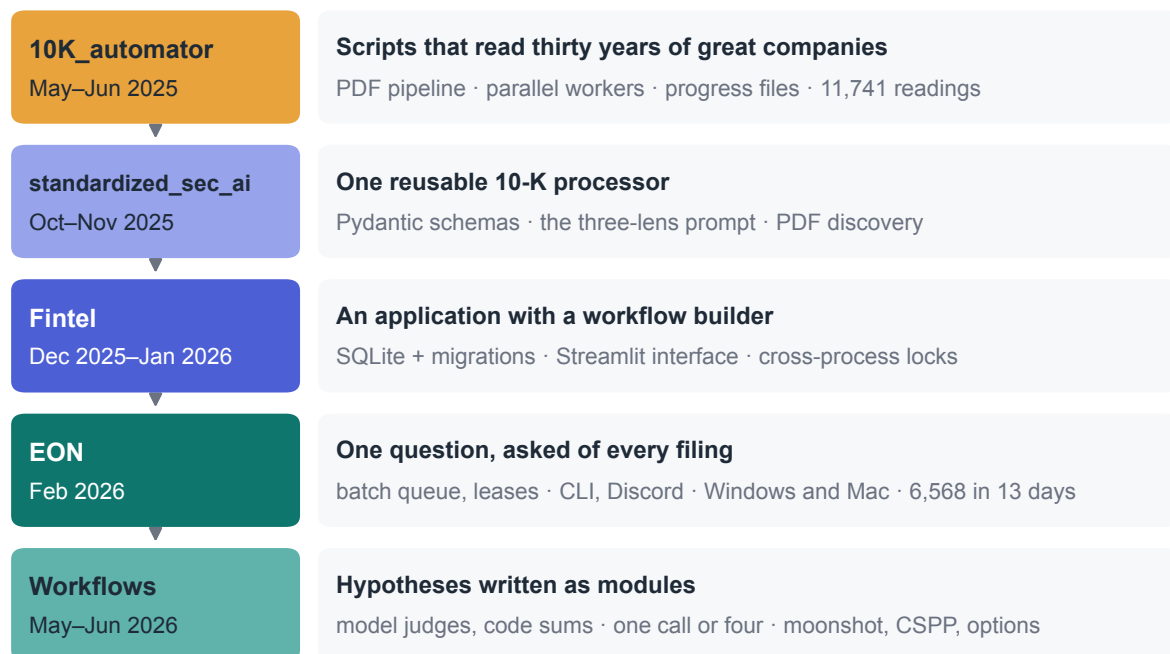


Figure 2: From scripts studying great companies to an application that reads every filing. Dates come from the origin project's files and the repository's git history.

So did two problems. The outputs had no fixed format; a script written later to flatten them found the six success factors spelled about 130 different ways. And the model did not know what day it was: of 1,919 comparative analyses run in May 2025, 1,916 dated themselves 2024. The saved outputs survived. More than a year later, their recorded dates made a different kind of test possible (Section 8.4), although local file timestamps are not proof of an unchanged archive.

3. From scripts to a processor

In October 2025 the scripts became a single reusable module, `standardized_sec_ai`. Its centre was `tenk_processor.py`, which downloaded, converted and read filings for any ticker, and its most important change was a single word: Pydantic.

Pydantic is a Python library for declaring the exact shape of data. Handed to Gemini as a response schema, it forces the model to answer in named fields of fixed types, and rejects any answer that does not fit. The project's own notes from that month celebrate it: "No more JSON parsing errors!" The 130 spellings of six factors could not happen again.

The same folder held `ppee.py`, the three-lens prompt from which EON's is still drawn: read the filing as Buffett would (moat, management, price), as Taleb would (fragility, tail risk, optional-ity), and as a contrarian would (what everyone believes, and why they might be wrong).

4. Fintel: an application

In December 2025 the processor grew into an application called Fintel, with a Streamlit web interface, a SQLite database and a visual workflow builder. A user could chain steps (pick companies, fetch filings, run a lens, filter, aggregate, export) and save the chain. The archive holds 16 of them: comparing three chip makers, reading three nuclear start-ups, pulling executive pay from a proxy statement.

The builder was good for discovering which questions mattered and bad for measuring anything. No two chains asked the same thing, so nothing accumulated. It was removed after three weeks.

What stayed was the infrastructure the builder had needed. Fintel was the first version that ran several analyses at once, and running several analyses at once broke things. Its first rate limiter used Python's `threading.Lock`, which only works inside one process. When the command line and the web interface ran together, or a batch spawned several processes, each had its own lock and the calls collided: the API answered with errors 429 and 503. The fix, recorded in a design note from January 2026, replaced it with a file lock (`portalocker`) that every process on the machine could see. The same month brought a nightly queue, an SEC rate limiter, and leases with heartbeats so that a crashed worker could not strand a company forever.

5. EON: one question, every filing

In February 2026 the project was renamed EON, the Erebus Observatory Network, and made one decisive change. Instead of many questions asked of a few companies, it would ask one question of every company: the three-lens prompt, with a 35-field schema, applied to every listed company's recent 10-Ks. Essays became a panel, the same measurement taken across 1,358 companies and six fiscal years. Everything in Part II depends on that.

Asking one question of every filing is easy to say. Figure 3 follows the request from the two interfaces through a shared service to a stored answer.

One reading, supported by a whole system

Solid arrows: filing → verdict / the surrounding machinery keeps it running

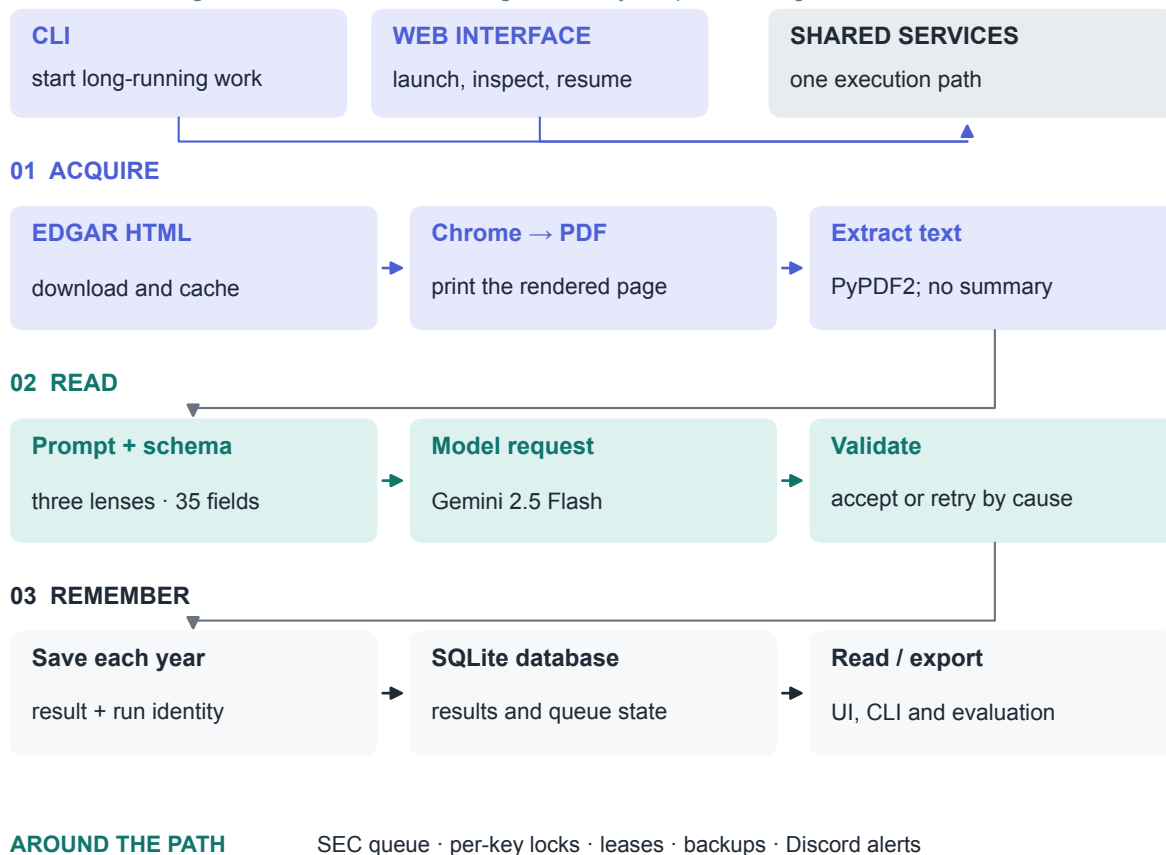


Figure 3: The CLI and web interface use shared services. Arrows trace the filing through acquisition, model reading and durable storage. The SEC queue, key locks, leases, backups and alerts support execution around that path. This is a schematic of responsibilities, not a literal call graph.

Each supporting component answers an operational constraint. The rest of this section takes them in turn.

5.1 Getting the words: why a PDF, and not XML?

The SEC publishes every filing on EDGAR, free, as HTML. The downloader is `sec-edgar-downloader`. EON routes requests through a shared queue, below the SEC's fair-access limit of ten requests a second: two seconds between requests, at most five at once, with a file lock so that parallel workers on one machine share the budget. Each filing is cached by ticker and year and never fetched twice; the cache now holds 27.5 GB of PDFs.

Why not simply read the HTML, or use the SEC's machine-readable XBRL data? The measurements answer both (Figure 4).

A 141-page filing becomes one word

The median 10-K on its way through EON / characters, logarithmic scale

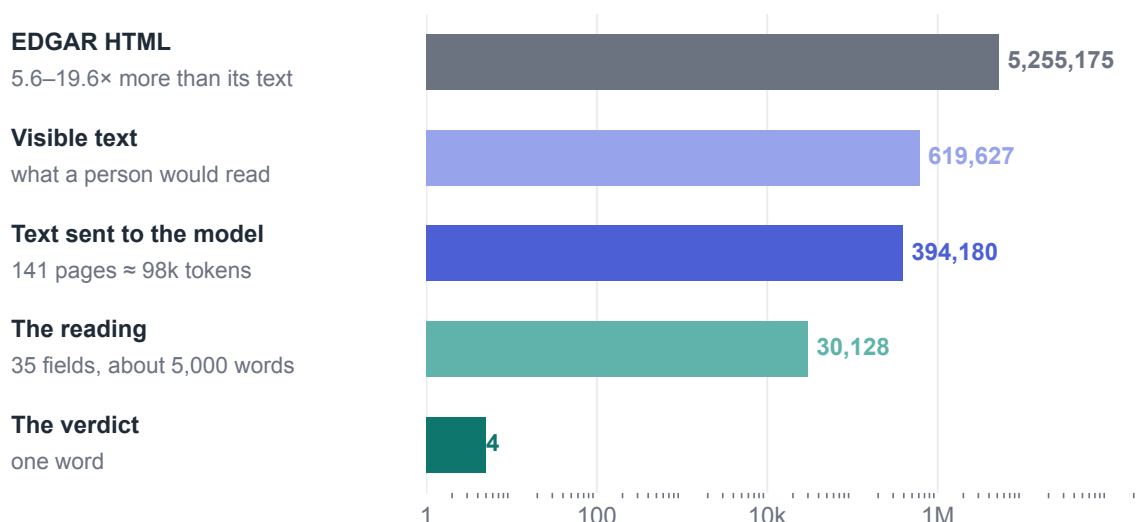


Figure 4: Characters at each stage for the median 10-K: raw EDGAR HTML across 17 sampled filings, their visible text, the text EON extracts from its PDF (40 sampled filings), the stored reading, and the verdict.

Raw EDGAR HTML is 5.6 to 19.6 times larger than the text a person would read. Most of the difference is markup, and much of that is inline XBRL: between 1,170 and 3,888 machine-readable tags per 10-K, plus a hidden header of tagged data. In Kraft Heinz's 2022 10-K, the invisible XBRL header (645,000 characters) is larger than the entire visible report (642,000). Feed that HTML to a model and most of what you pay for is markup; strip the tags naively and the hidden data pours into the text.

XBRL itself is excellent for numbers: revenue, debt, cash flow, tagged and comparable. But EON's questions are about judgement: moats, fragility, what management says about risk. That lives in the narrative, which XBRL does not structure.

So EON prints each filing the way a person would see it. A headless Chrome browser opens the HTML, waits between 3 and 15 seconds depending on the file's size so that the page finishes rendering, and calls Chrome's own print-to-PDF command. The result is exactly what a reader sees: tables laid out, hidden XBRL gone, 141 pages at the median. PyPDF2 then extracts the text in about 6 seconds. Nothing is summarised or cut; a filing too long for the model's context is skipped, not truncated.

The route costs a browser per filing, and browsers misbehave. A conversion that hangs is abandoned after 180 seconds and the browser restarted. Orphaned Chrome processes accumulate over a long batch, so EON sweeps them up every 50 companies, or sooner if memory use passes 80%. Tables and layout can still be mangled in extraction; that path has not been audited, and parsing HTML directly would be cleaner. For a machine built to read like a person, printing the page like a person was the simplest reliable answer.

5.2 Asking the question

The median filing arrives as about 394,000 characters, roughly 100,000 tokens. EON appends all of it to the three-lens prompt and sends one request to Gemini 2.5 Flash, with the 35-field schema attached. The model returns about 30,000 characters, 5,000 words of analysis, and one verdict (Figure 4).

Pydantic validates every response before it is stored. A response that does not fit is not a reading. Failures are sorted by cause before they are retried: a rate limit waits for exactly as long as the API's own retry-after field says; a transient server error backs off and tries again; a context overflow is marked skipped and never retried. Across all 6,973 three-lens readings, the model's input came to roughly 0.7 billion tokens.

5.3 Twenty requests a day

The binding constraint was not compute, storage or money. It was requests. The February configuration rotated 25 API keys, each assigned a local budget of 20 requests a day, with a mandatory 65-second pause between requests on any one key. That is 500 filings a day in the local scheduler.

This describes the implementation, not an entitlement to multiply provider quota. Google's rate-limit documentation applies limits per project, not per API key (Google, 2026). The retained records do not establish the keys' project allocation or whether the historical arrangement was permitted. The paper therefore makes no claim of provider approval; a new deployment must use its project's actual quota.

Every other design decision in EON follows from that number, and getting it right took three attempts.

The first, in Fintel, was the in-process lock that different processes could not see. The second was the file lock that fixed it: one lock for the whole machine, held through each request and its 65-second pause. It worked, and it made 25 keys exactly as slow as one. The third, in EON, gave every key its own lock file and added a 25-slot semaphore within each process. Requests on different keys could overlap; each key's file lock coordinated its use across processes on the same machine. The pause adapts: it shrinks after a run of successes, down to about 21 seconds, and grows by half after a rate-limit error.

A fourth bug was quieter. On 6 February 2026 a commit titled "Remove double-counting of API key usage that halved effective capacity" found that every request was recorded twice, once by the model provider and again by seven of its callers, so each key reached its limit of 20 after 10 real requests. For a while the machine had been running at half speed and reporting itself full.

Parallelism allowed the scheduler to use its configured daily budget. Figure 5 shows the difference between the two ways the command line runs.

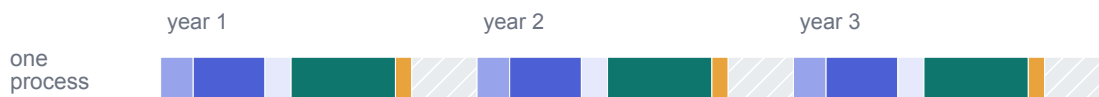
The market-wide batch ran from 8 to 21 February 2026 and read 6,568 filings from 1,327 companies (Figure 6). At its busiest it finished 189 readings in an hour, one every 19 seconds across all workers. But it was active for only 62 of the thirteen days' hours. For about 80% of its life, the fastest part of the machine was waiting for midnight.

Of the 1,410 companies attempted, 83 failed: 61 because no filing could be retrieved, and 22 after the model call failed repeatedly.

One company runs in a line; a batch runs in lanes

How the CLI executes / schematic, not to scale

A eon analyze AAPL --years 3



B eon batch companies.csv (25 keys; five lanes shown)

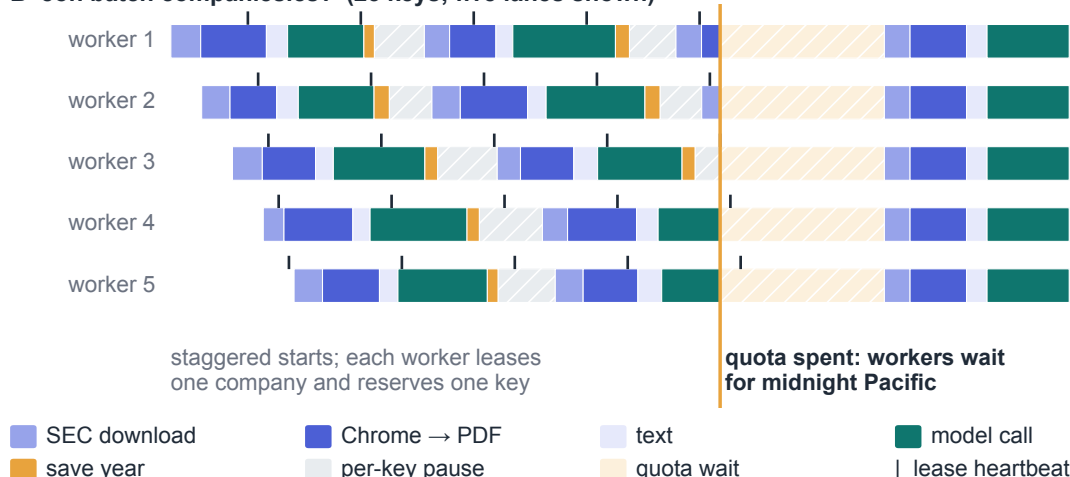


Figure 5: A single analysis downloads, prints, extracts, asks and saves one fiscal year after another. A batch starts up to 25 workers with staggered starts; each leases one company, reserves one key, and renews its lease with a heartbeat. When every key is spent, all workers wait for the reset at midnight Pacific. Schematic.

5.4 The database is the batch's memory

A batch that runs for two weeks will be interrupted: by a crash, a reboot, a power cut, a closed laptop. The database is what lets it pick up where it left off, preserving completed years and reducing repeated work (Figure 7).

EON uses SQLite, a single file, because the whole system runs on one machine and a server database would add a service to install, configure and keep alive. SQLite's weakness is that only one writer can write at a time. EON works with that rather than against it.

- **Write-ahead logging.** WAL mode allows readers to overlap the writer, so the web interface can browse results while a batch writes new ones.
- **Patience, then retries.** Each connection waits up to 30 seconds for the lock, with up to ten attempts in total and exponential backoff plus random jitter, so that competing workers do not retry in lockstep.
- **Leases.** A worker claims a company with a conditional update (set it to running, but only if it is still pending), so exactly one worker wins. The claim carries a lease of three hours, renewed by a heartbeat every five minutes. If a worker dies, its heartbeat stops, its lease expires, and the company returns to the queue.
- **Crash recovery.** On start-up EON checks whether the process that last held the batch is still alive. If not, its companies go back to pending.

The quota is the clock

Market-wide batch, 8 to 21 February 2026 / readings per hour, UTC

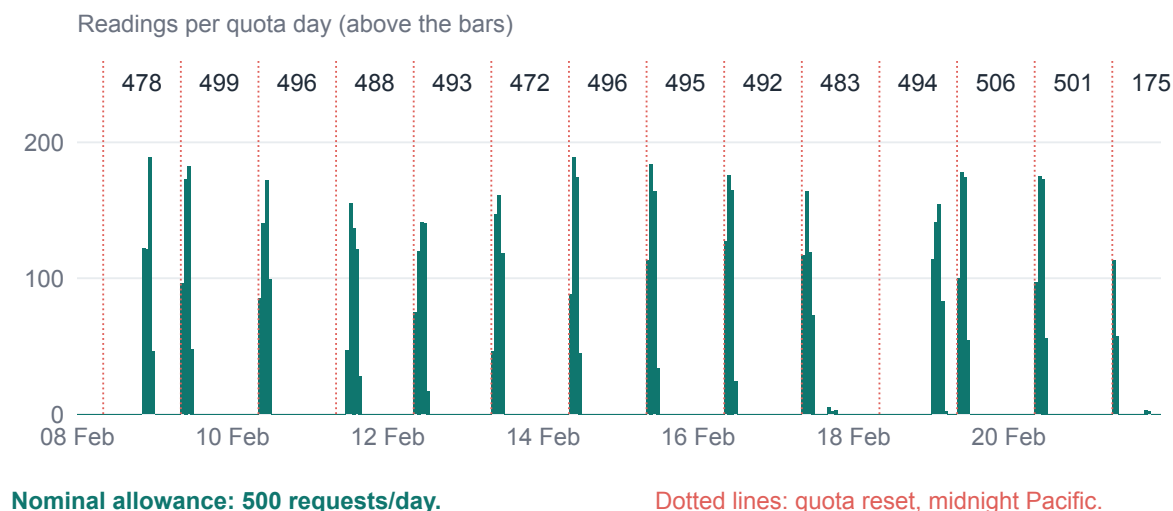


Figure 6: Readings per hour during the February batch, with approximate quota-day totals of 472 to 506 above the bars. The reset boundary is reconstructed, so a day can slightly exceed 500.

- **Per-year checkpoints.** Each fiscal year's reading is saved the moment it arrives. An interrupted company resumes from its last completed year, not from the beginning.
- **A unique index.** In February a double-save pattern (once as each year finished, once at the end) quietly stored some readings twice; `INSERT OR IGNORE` had nothing to ignore against. Migration 14 removed the duplicates and added a unique index. (Separate batches that overlapped still read some company-years twice, which is why 6,973 stored rows hold 6,653 distinct company-years; the evaluation keeps the first.)
- **Backups.** During a batch, EON copies the database with SQLite's own backup API and keeps the last five.

The schema itself has been through 14 migrations, each written to be safe to run twice and applied automatically at start-up. One earlier file in the repository's history is named `fintel.db.corrupted`: a reminder of why the rest exists.

5.5 Interfaces and alerting

EON has a command line and a web interface, and they do different jobs.

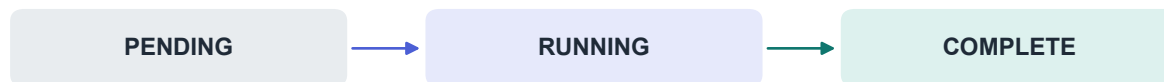
The command line is for long work. A batch of a thousand companies over ten years needs twenty days of quota; the README's advice is to start it in a terminal multiplexer and detach, because a batch must outlive any browser tab. `eon batch`, `eon analyze`, `eon export` and `eon scan-contrarian` are built with Click.

The Streamlit web interface is for looking: launching a single analysis, browsing history, reading results, managing the batch queue, and resuming anything interrupted. For a while the two had their own code paths and drifted apart. On 6 February both were rebuilt on one shared service layer, so a batch started from either behaves the same way, and a batch started from the command line can be watched from the browser.

Many workers. One writer. A durable memory.

SQLite on one machine / claiming work and saving results are separate operations

A CLAIM THE WORK



Conditional UPDATE: only a pending item can be claimed.

Heartbeat renews a 3 h lease every 5 min; stale work returns to pending.

B SERIALIZE THE WRITES



Busy writer? Wait up to 30 s; retry with backoff and jitter (10 attempts total).

C KEEP TWO KINDS OF MEMORY



Per-year checkpoints reduce repeated work. A unique index rejects duplicate rows.

SQLite backup API keeps five copies. Leases are not an exactly-once guarantee.

Figure 7: A conditional update claims a pending company; heartbeats maintain its lease. SQLite serializes short write transactions while WAL allows readers to overlap the writer. Batch tables remember progress; analysis tables preserve readings. Checkpoints and uniqueness constraints reduce repeated work but do not promise exactly-once model calls.

Discord provides alerts. A batch that runs overnight for two weeks needs a way to say something while its owner is asleep or away. EON posts to a Discord webhook when a batch completes or fails, when every key is spent and it is waiting for the reset, and when disk space runs low: the 27.5 GB of PDFs are a real risk. Before starting, it checks that there is room.

5.6 Cross-platform execution

EON ran on two computers. The February batch ran on Windows; the database still stores its paths with backslashes. Development happened on a Mac, which kept its own database of 1,184 readings. A pull request titled “multi-os support”, merged on 2 February 2026, made the difference invisible. File locks use `portalocker`, which speaks both Windows and Unix locking. Paths are built with the platform’s own separators. Stray Chrome processes are cleaned up with `kill` on the Mac and `taskkill` on Windows.

5.7 Is it efficient?

It depends on what is scarce.

Against the quota, EON aims to make every request count: each reads a whole filing, and checkpoints reduce repeated calls after an interruption. Against the clock, it is not: the batch spent about 80% of its time waiting. Against compute, it is wasteful but cheap: a browser per filing and about 6 seconds of extraction buy a readable text whose fidelity still needs an audit. Against storage, it is heavy: 27.5 GB of PDFs for a database of 240 MB.

If requests were plentiful, the design would change. Parsing HTML directly would drop the browser. Caching extracted text instead of PDFs would cut the disk. Sending only the sections each lens needs would cut tokens substantially. Section selection trades coverage for speed; direct HTML parsing and text caching could improve efficiency without that trade, provided their output was checked. The quota made those optimisations less urgent.

6. Workflows as hypotheses

With the panel built, the questions became narrower. From May 2026 new analyses were written as custom workflows: Python files, discovered automatically, each with its own prompt and schema. They became a way to write down a hypothesis and test it on hundreds of filings.

CSPP turned a long analytical framework (17 scored components in 5 domains) into a reading. Its first version asked for everything in one call, with a schema of about 106 KB, 28 times the size of one lens. When any of its roughly 200 required fields drifted, validation failed and the whole reading was lost. On 20 May 2026 it was split into four smaller calls, merged in code, with a partial result kept if one part fails. The rewrite also stopped trusting the model's arithmetic: it supplies component scores, and Python adds them up.

Moonshot × **Excellence** asked whether a company offers an asymmetric bet and whether it resembles the great companies of 2025. It uses one call on purpose: it can include up to 250,000 tokens of reference material, and a second call would resend all of it and spend another request. It rates six dimensions on anchored 0–10 scales and computes the totals in code. It also carries a watchdog, added after a large filing hung a call for about two hours without an error.

Contrarian Hunter and the **options scans** brought the 2025 experiments into EON. The February 2026 options scan, Asymmetric Options V4, read the latest filing of 1,415 companies in two days and assigned each a directional bias. Section 8.5 tests it.

Two rules came out of this work. Split a question the model cannot answer reliably in one piece. And let the model judge while code does the counting.

Part II · What it read



Illustration — The reading desk. An archive, a report and a recorded judgement.

7. Every question it was asked

Across five versions, the model gave 29,375 stored answers (Figure 8).

Table 1: Stored answers by analysis family.

Analysis	When	Answers	Asked
Excellence study	May 2025	15,632	What made great companies great, and who resembles them?
Contrarian scan	May 2025	1,933	Where is strength hidden that others miss?
Options ideas	June 2025	1,933	Buy calls, buy puts, or neither?
Fintel analyses	Dec 2025–Jan 2026	286	Fundamentals, single lenses, syntheses
Three-lens verdicts	Jan–Jul 2026	8,157	Buy, hold or sell, as Buffett, Taleb and a contrarian
Asymmetric Options V4	Feb 2026	1,415	Which way is the tail risk?
CSPP and Moonshot	May–Jun 2026	19	Frameworks too large for one call

Read together, the answers show a model with habits.

- **It likes to hold.** Of 6,411 dated three-lens verdicts, 50.8% say HOLD, 27.5% BUY, 16.5% SELL, 3.8% STRONG BUY and 1.4% STRONG SELL. The mix barely moves from year to year, through a boom, a bear market and a recovery. Conviction is Medium 4,632 times, High 1,516 times and Low only 148 times.
- **It finds danger everywhere when asked to look for it.** Asked where the tail risk lies, the February 2026 options scan gave 873 companies (62%) a put bias and 27 (2%) a call bias.

29,375 stored answers

Archive counts by analysis family / labels give exact totals

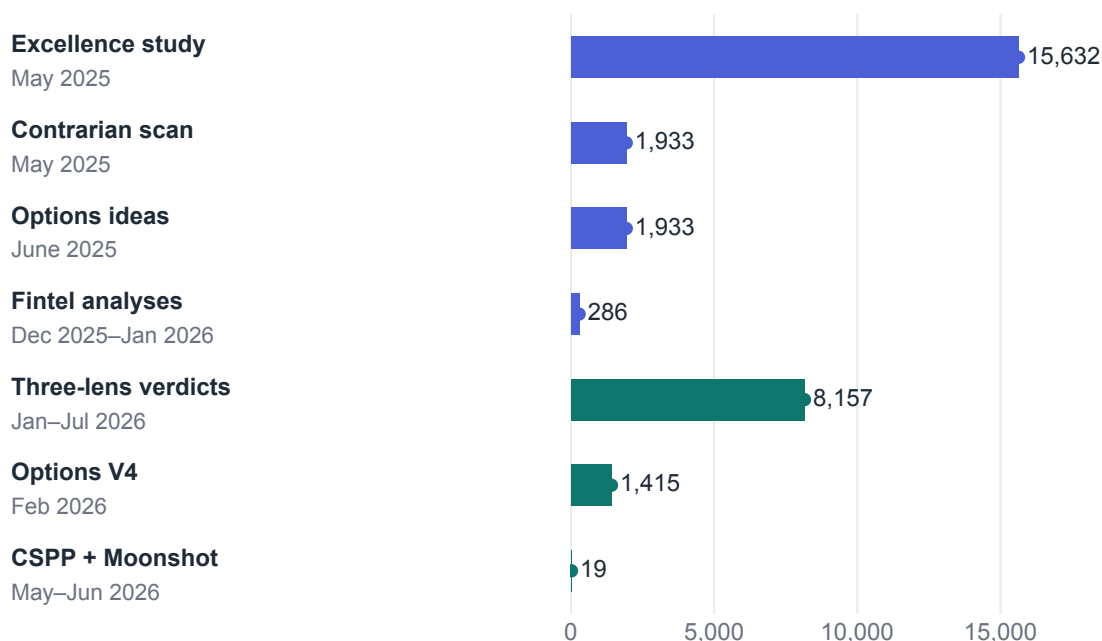


Figure 8: Stored answers by analysis family; bar length represents count, with exact totals beside each bar.

- **It favours the famous.** The 2025 contrarian scan, meant to find hidden gems, put MicroStrategy first, followed by Apple, AST SpaceMobile, Tesla, Microsoft and Nvidia. Only 27 of 1,933 companies scored 70 or more.
- **Its numbers cluster.** A quarter of all 2025 compounder scores are exactly 68, and three values cover 44% of companies.

None of this says whether the answers were right. That takes a test, and the test is harder than it looks.

8. Was it right?

8.1 Three dates

In February 2026, while the market-wide batch was still running, I checked the readings against stock returns. BUY-rated filings had beaten SELL-rated ones by 14.1 percentage points over two years. The report concluded that the model had genuine stock-picking skill.

It had skipped a question. The model was being graded on years its training data could already cover. Gemini 2.5 Flash's training data ends in January 2025, according to Google, and this paper uses 31 January 2025 as the line. Show it a 2021 annual report and ask whether to buy: the report ends in 2021, but the model may know how 2022 went. A backtest like that is a trader handed tomorrow's newspaper. Their record would be extraordinary, and it would say nothing about their judgement.

Every verdict therefore has three dates: when the company filed, where the model's knowledge ends, and when the verdict was written. Their order decides what a test can prove (Figure 9).

Three dates decide what a test can prove

When the filing appeared, where the model's knowledge ends, and when the verdict was written

WITH HINDSIGHT



The model may already know how the story ended.

AFTER THE CUTOFF



The filing follows the stated cutoff; the verdict still came late.

WRITTEN IN ADVANCE



The verdict exists before anything it predicts. Only this is a forecast.

Figure 9: Only the third arrangement, a verdict recorded before anything it predicts, is a forecast in the ordinary sense. Event order is schematic.

Every test below uses one reading per company and year; entry at the first close after the filing (or, for the ledgers, after the reading); returns in excess of SPY, the S&P 500 fund; and significance from shuffling verdicts 10,000 times within each year, and within each year and industry. Appendix A has the details.

8.2 With hindsight, and after the cutoff

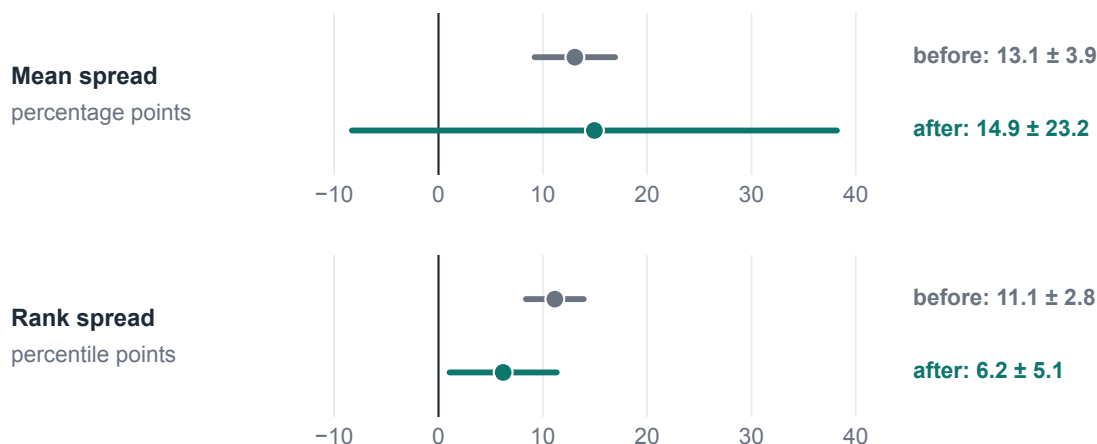
The primary post-cutoff test was the six-month mean spread, specified before the evaluation: 4.6 points, $p = 0.094$. It did not pass the conventional 5% threshold. The one-year horizon was added in a later amendment; the rank statistic was added after inspecting the outlier and is a robustness check. The comparisons below explain the findings without replacing that primary test.

For one-year windows that closed before the cutoff, 1,099 BUY readings beat SPY by 2.2 points and 657 SELL readings trailed it by 10.9: a spread of 13.1 percentage points ($p \leq 0.0001$).

For the 1,443 filings published after the stated cutoff, the same test gives a different picture (Figure 10). Ranked by where each stock finished within its year, the spread halves, from 11.1 to 6.2 percentile points. The average spread, 14.9 points, looks unchanged, but its uncertainty is ± 23 points.

After the cutoff, ranks halve; averages turn to noise

One-year BUY minus SELL, the same test on both sides of the cutoff / 95% intervals



Before: 1,099 BUY, 657 SELL. After: 339 BUY, 201 SELL.

Figure 10: One-year BUY-minus-SELL mean and rank spreads on aligned zero axes, with descriptive 95% intervals; the post-cutoff one-year test was added after the primary six-month test.

Where the edge lived also changed (Figure 11). Before the cutoff, the five verdicts were broadly ordered: STRONG SELL companies finished at the 43rd percentile of their year, SELL at the 43rd, HOLD at the 50th, BUY at the 54th, STRONG BUY at the 60th. After it, BUY still finished above the middle, but SELL rose to the 49th percentile, level with HOLD. The SELL calls, which had carried most of the historical spread, stopped working.

A model that remembered which companies collapsed would look like this. So would a model whose caution suited 2021 to 2024 and not 2025. The data cannot choose between them. Survivorship also matters: selecting companies still listed in 2026 omits failures and can lift the measured returns of SELL-rated companies. That biases this comparison toward the apparent loss of SELL performance; it cannot be attributed solely to the model.

The post-cutoff sample retains a small rank association. By rank, BUY readings beat SELL readings by **5.9 percentile points** ($p = 0.0059$) in the sample eligible for within-industry shuffling. Figure 10's 6.2-point spread uses all eligible one-year readings; the 5.9-point statistic drops singleton industry-year blocks and recomputes the within-year ranks. That statistic was chosen after the next figure was seen, and it rests largely on one filing season. By average, the result is inconclusive: 4.6 points at six months ($p = 0.094$) and 14.9 at a year ($p = 0.215$).

After the cutoff, the SELL calls stop working

Average one-year return percentile by verdict / 0.5 is the middle of the pack

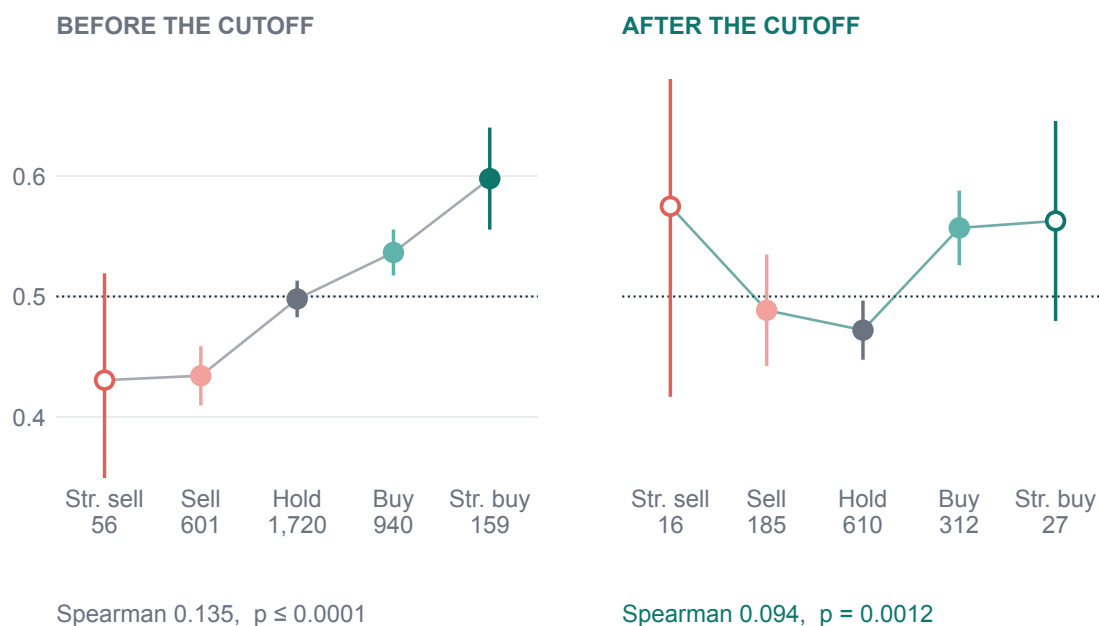


Figure 11: Average within-year percentile of one-year excess return by verdict, ± 1.96 standard errors; hollow points have fewer than 60 readings. The verdict order still correlates with outcomes after the cutoff (Spearman 0.094, $p = 0.0012$), driven by BUY rather than SELL. Descriptive.

8.3 Outlier sensitivity and company-level outcomes

The average fails because of one company (Figure 12). Babcock & Wilcox filed its 10-K on 31 March 2025, closed at \$0.46 the next day, and stood at \$15.72 a year later: 3,314 points ahead of SPY. That single reading adds 10 points to the BUY group's average. The price series was checked against Nasdaq's independent record; there was no split. EON read the filing on 9 February 2026, rated it BUY, and by then the stock had already risen about twentyfold. The documented cutoff precedes that rise, and the call path has web search switched off. Those facts reduce one route to hindsight; they do not turn a late reading into a forecast.

One stock can decide a mean

Post-cutoff filings / one-year excess over SPY, percentage points / symmetric log axis

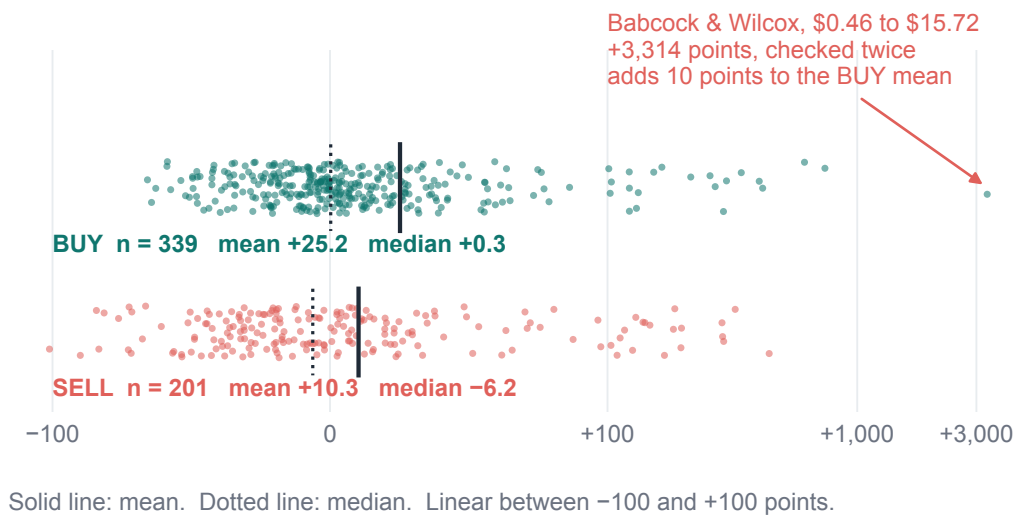


Figure 12: Every post-cutoff BUY and SELL reading with a one-year outcome. Solid lines are means, dotted lines medians; the axis is linear to ± 100 points and logarithmic beyond.

Averages also hide what the verdicts look like one company at a time. Figure 13 shows 24 companies drawn at random, with a published seed, from the 195 that EON read in every year from 2020 to 2024 and rated both BUY and SELL at least once. Of their 70 BUY and SELL calls, 40 went the right way: 57%. This combines hindsight-exposed and post-cutoff windows; it is a descriptive illustration, not a forward hit rate.

Company by company, the edge is thin

24 companies drawn at random, seed 20260930 / one-year excess return over SPY, points

		FY2020	FY2021	FY2022	FY2023	cutoff	FY2024
ACIW	ACI Worldwide	-27 ✓	-17	-8	+54		-44 X
BW	Babcock & Wilcox Inc	-9	-19 ✓	-118	+4		+3,314 ✓
COKE	Coca-Cola Consolidated	+80 X	+32 ✓	+24 ✓	+53 ✓		+26 ✓
CTBI	Community Trust Bancorp	-9 X	+12	-35	+23 ✓		-2 ✓
DGX	Quest Diagnostics	+1 ✓	+14 ✓	-39 X	+21 ✓		+8 X
EME	EMCOR Group	+4	+52 ✓	+38 X	+9 ✓		+64 ✓
EVRG	Evergy	+6 ✓	+5 X	-43	+28		+12
EYPT	EyePoint	-2	-75	+724	-79 X		+98 X
FBNC	First Bancorp	-2 X	+2 ✓	-48	+6 X		+25 X
FMBH	First Mid Bancshares	-12 X	-17	-25 ✓	+8		-7 X
GNTX	Gentex	-27 X	+2 ✓	+1	-50		-19 ✓
IEP	Icahn	-11 X	+22 X	-86 ✓	-54		-19
KTOS	Kratos Defense & Security Solutions	-49	-9	+14 X	+6 ✓		+227
MGRC	McGrath RentCorp	-16 X	+38 X	-9 X	-17		-24
MPB	Mid Penn Bancorp	-8 X	+9 X	-56 X	+25 ✓		+5 ✓
MTZ	MasTec	-25	+38 ✓	-35 ✓	+22 X		+119
MYRG	MYR Group	+41 ✓	+36	+18	-41		+109 X
NIC	Nicolet Bankshares	+11 ✓	-13 ✓	-23	+34		+15
NWBI	Northwest Bancshares	-9 X	+12	-41 ✓	-1		-10
OC	Owens Corning	+8 ✓	+9 ✓	+18 ✓	+1		-39 ✓
PRLB	Proto Labs	-80 ✓	-37	-16	-10		+38 ✓
UVE	Universal Insurance Holdings	-31 ✓	+82 X	-20 X	-4		+50
VLV	Valley National Bancorp	+1 ✓	-5 ✓	-60	+6		+18
ZD	Ziff Davis	-13	-17	-43	-60 ✓		-54 X

Strong buy Buy Hold Sell Strong sell ✓ 40 of 70 calls

Figure 13: Colour is the verdict; the number is the one-year return over SPY in percentage points; a tick marks a BUY that beat SPY or a SELL that trailed it. The rule selects on verdicts, never outcomes. Because the median stock trailed SPY in these years, SELL ticks come cheaply. Mixed hindsight and post-cutoff windows; descriptive only.

8.4 A ledger written in 2025

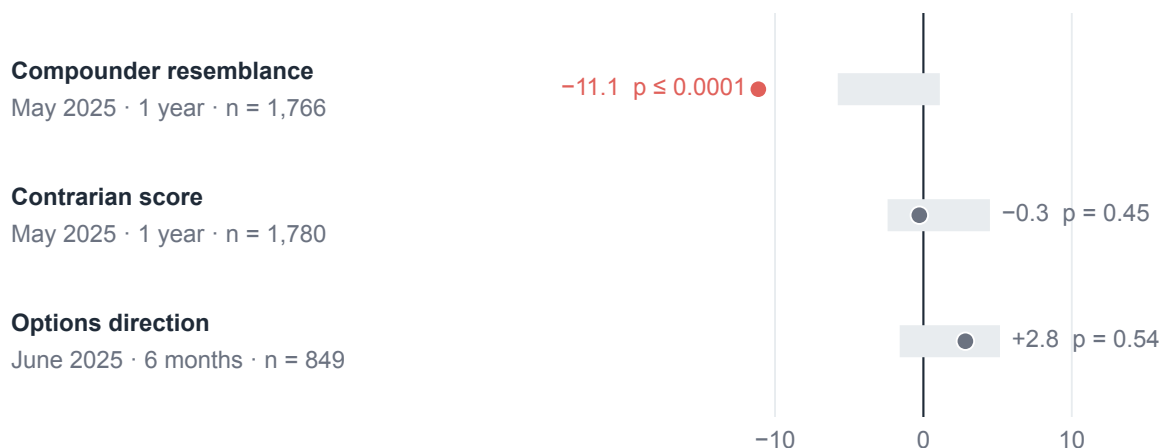
The strictest evidence is the oldest. The 2025 scripts wrote their scores to disk in May and June 2025, after the model's cutoff and before any of the returns they might predict. They are a ledger, and they need no argument about training data (Figure 14). Their dates come from file timestamps, not a sealed archive, and I had looked at the first result before writing the test.

Resemblance to great companies pointed the wrong way. Across 1,766 companies, the higher the 2025 compounder score, the worse the next year: Spearman -0.14 , $p \leq 0.0001$. The top fifth trailed SPY by 17.8 points on average; the bottom fifth beat it by 18.6. A sensible idea, carefully applied, failed its first year. One plausible reason, untested, is that looking like a past winner was already in the price. The original ambition was about decades, which one year cannot settle.

The contrarian score and the options calls showed nothing. Spearman $+0.01$ ($p = 0.71$) for the contrarian score. This correlation differs from Figure 14's -0.3 -point top-minus-bottom quintile rank spread; both are near zero. The options direction contrast was calls minus puts $+2.8$ percentile points ($p = 0.54$) over six months for the options ideas.

Looking like a great company was a bad sign

Scores written down in 2025, before their outcomes / top minus bottom, return percentile



Band: middle 95% of industry shuffles, not a confidence interval.

Figure 14: Top fifth minus bottom fifth of each 2025 score (calls minus puts for options), in percentile points of return, against the middle 95% of 10,000 industry-preserving shuffles.

These are not EON's verdicts. Across 1,054 companies scored by both, the compounder score and EON's verdict are almost unrelated (Spearman 0.05).

8.5 A ledger written in February 2026

EON left a ledger of its own. The Asymmetric Options V4 scan of 25–26 February 2026 recorded, for 1,415 companies, a directional bias and an asymmetry score, six months before the outcomes it would be graded on. I wrote the test before running it (Figure 15). Of the 1,415 scanned

companies, 1,365 had usable six-month outcomes. The 873 put-bias names in the full scan become 847 in this graded subset.

Magnitude association; no directional evidence

Options scan of 25–26 February 2026, graded six months on / written before the outcome

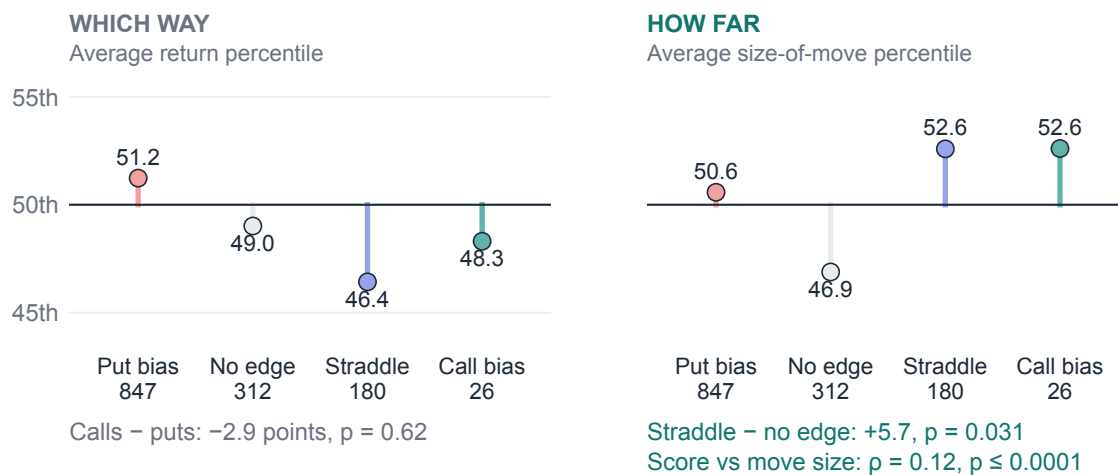


Figure 15: Six-month return and absolute-return percentiles for 1,365 companies; the magnitude association has no trailing-volatility baseline.

The directional test found no evidence of skill. Stocks marked for calls did no better than stocks marked for puts (-2.9 percentile points, $p = 0.62$); put-bias stocks in fact beat SPY slightly more often than the rest. Its asymmetry score correlated with the size of the move, up or down (Spearman 0.12, $p \leq 0.0001$), and stocks marked “straddle”, the scan’s way of saying something big will happen, moved more than those marked “no edge” ($p = 0.031$).

This is an association with absolute excess returns, not a test of realised volatility or profitable options trading. Trailing volatility, company size and sector could explain it; no baseline using those variables was fitted. The scan therefore has not demonstrated incremental predictive skill. The hypothesis that filings expose binary risks remains open.

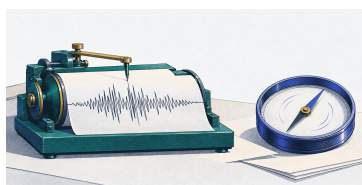


Illustration — A recorder and a compass.

8.6 What the tests say together

The primary post-cutoff test was inconclusive. The later rank association, the failed compounder score and the unbenchmarked magnitude association describe different tests; none substitutes for a prospective trading record. The evidence has limits that apply throughout: the later tests cover roughly the same eighteen months; the universe was drawn from companies still listed in 2026; industry is controlled but size, value, quality and momentum are not; extraction and the model’s facts are unaudited; and returns are not trades. Section 11 is what the paper does about that.

Part III · What it taught



Illustration — The unattended lectern. A calendar, repeated reports and a selective view.

9. How the model went wrong

The most useful findings in this project are not about stocks. They are about how a fluent language model fails when it is asked to do a job thousands of times (Figure 16).

It did not know what day it was. Run in May 2025, the model dated 1,916 of 1,919 analyses 2024, a year behind, and proposed 91 option expiries that had already passed. It reasons from its training-time present unless told otherwise. EON's later workflows stamp dates in code.

It may remember the future. A model trained after the events it is asked to judge can borrow from them without saying so. A search of every reading for terms with known first-use dates found no careless leaks: the Inflation Reduction Act appears in 300 readings, never in one of a filing written before the law. But a verdict nudged by memory leaves no trace in its prose.

Its numbers are less precise than they look. A quarter of all compounder scores were exactly 68. A 0–100 scale that behaves like a four-point one invites false precision. The later workflows use anchored scales, each defined at 0, 5 and 10.

It leans. It holds half the time. Asked to look for tail risk, it found it in 62% of companies and upside in 2%. The framing of a question shapes the answer's direction as much as the evidence does.

It favours platitudes and the famous. It credited “continuous innovation” to every great company, and its contrarian scan's hidden gems included Apple, Microsoft and Nvidia.

It drifts in format. Without a schema, six factors took 130 spellings. With one, 462 verdicts still opened with a label (“Overall investment recommendation: ...”) instead of the answer, and conviction arrived in half a dozen phrasings. Every parser must expect this.

It cannot always add. CSPP stopped trusting its arithmetic and recomputed every total in code.

Eight ways a fluent model went wrong

Measured in the project's own outputs, 2025–2026



Figure 16: Each number is measured in the project's own stored outputs.

It sometimes stops. One large filing hung a call for about two hours without an error. Every call now has a watchdog.

It values without prices. Apple's SELL rests on a "highly stretched valuation", but a 10-K contains no current share price. The model supplies one from memory, which is exactly the channel through which hindsight can leak.

10. Why it was worth doing

EON tested what one builder could learn from public filings with a language model and a constrained request budget. Competing with institutional trading systems was never its practical objective. The value was in making the question precise enough to expose a disappointing answer.

It was three projects at once.

As a finance project, it turned a vague question (can reading annual reports well tell you anything?) into a measurable one, and taught the details that make such measurements honest: excess returns and their benchmark, industry composition, survivorship, the difference between an average and a rank, one stock deciding a mean, and above all look-ahead. The primary test was inconclusive, and the remaining associations need stronger controls. Learning to distinguish those statements was part of the result.

As an AI project, it measured a language model at scale, on the same task, thousands of times, and caught it misdating its work, drifting in format, clustering its scores, leaning bearish

when asked about risk, and possibly remembering the future. These are observed failures of this model and these prompts. Repeated measurement made them visible; it does not establish their prevalence across all language models.

As a systems project, it forced real engineering under a hard constraint: cross-process locks, key rotation, leases and heartbeats, crash recovery, a database built to survive two weeks of interruption, two operating systems, two interfaces, and an alarm that works while you sleep. None of that was planned. Every piece answers a failure that happened.

And it left something no one else has: 29,375 answers written down with their dates, a method for testing them honestly, and a bet.

11. The prospective ledger

Every lesson in this paper points to one practice: write the verdict down, with its date, before the future arrives. So the paper makes a forecast of its own (Figure 17).

A prospective test of 1,257 frozen verdicts

49 extreme verdicts shown / primary test: all 371 BUY-type versus 252 SELL-type

STRONG BUY · 30

AD	AMG	AMPH	BBW	BCRX	CB	CNK	CRK	EME	GRBK
IBKR	INVA	IRDM	KFY	LNG	NEU	NXST	PEGA	PFS	POWL
QCRH	RNR	SBH	SCSC	SRCE	SYF	USLM	VSH	WINA	WLKP

STRONG SELL · 19

AAOI	ALX	ARDX	AVAV	CELH	CSR	CWST	DCOM	INDV	NUTX
OGS	ORC	PLUG	RGEN	RIG	SMR	UHAL	UNIT	ZBRA	

Frozen 28 September 2026 · Research ledger, not recommendations · SHA-256 b5a03624d885c0c9...

Figure 17: All 49 strongest verdicts are shown; the pre-specified test covers all 371 BUY-type against all 252 SELL-type calls in the 1,257-company ledger.

The ledger holds EON's most recent verdict for each of 1,257 companies: 634 HOLD, 341 BUY, 233 SELL, 30 STRONG BUY and 19 STRONG SELL. The whole file is identified by its SHA-256 fingerprint, which begins b5a03624d885c0c9. A hash detects a changed file; it does not independently establish when the file was frozen. Public timestamped release is still needed for that. The test was chosen in advance: BUY-type minus SELL-type percentile rank of one-year return over SPY, from the first close after 30 September 2026. Many of these verdicts are already a year old, since they read filings from early 2025; that is part of the test.

The next vintage should do better than a frozen file on one laptop. Read the fiscal 2025 reports now, with the same model and prompt; store each reading with the filing's identifier, a hash of the text the model saw, the prompt version and the time; pin the model, because a new model

has a new cutoff and turns clean evidence back into hindsight; tell it the date; and publish the fingerprint somewhere public before the returns arrive.

12. Conclusion

I set out to build a patient reader. It took five versions and seventeen months, and most of the work was not reading at all. It was waiting for midnight, printing web pages, keeping two dozen workers from colliding, and making sure nothing was lost when something broke.

The system produced structured readings at scale. The primary post-cutoff six-month test did not establish a return spread (4.6 points, $p = 0.094$). The positive rank result was post hoc; the options magnitude association lacks a volatility baseline. Neither establishes a tradable advantage.

The project nevertheless leaves a reproducible system, a record of model failures, and a sharper experimental question. A forecast earns its name when it is recorded before the events it predicts. The frozen ledger makes that next test explicit.



Illustration — Open October 2027. The next evaluation date.

Appendix A: Methods

Scope. One reading per ticker and fiscal year, the first stored. Of 6,653 such readings, 6,651 were made with Gemini 2.5 Flash; two made with Gemini 3.5 Flash are excluded. A usable observation needs a parsed verdict, a filing date and prices: 6,203 readings of 1,309 companies.

Sample reconciliation. The following populations answer different questions; archive totals and batch subsets are not successive exclusions from one funnel.

Table 2: Sample populations and evaluation filters.

Population	Readings	Companies	Meaning
Main EON three-lens archive	6,973	1,358	Stored answers, including repeated company-years
Main archive plus Mac archive	8,157	—	6,973 + 1,184; archive count, not deduplicated
February market-wide batch	6,568	1,327	One batch within the main archive
First stored company-year, all models	6,653	—	Repeated company-years removed
Gemini 2.5 Flash only	6,651	—	Two other-model readings excluded
Parsed verdict and filing date	6,411	—	Both fields present; no price requirement
Usable entry-price sample	6,203	1,309	Parsed verdict, date and entry price; exit availability varies by horizon
Prospective ledger	1,257	1,257	One latest eligible verdict per company under the separate ledger protocol

A dash means the company count is not reported for that stage. The parser alone accepts 6,519 readings and the date check alone accepts 6,543; their intersection is 6,411. Horizon-specific samples are smaller still.

Returns. Entry at the first close strictly after the filing date. Excess return is the change in adjusted close over 126, 252 or 504 trading sessions minus SPY's change over the same dates, in percentage points, not annualised. For the ledgers, entry follows the recorded reading date. Prices are a frozen Yahoo Finance snapshot through 28 September 2026; the largest outlier was checked against Nasdaq.

Samples. “Before the cutoff” means the return window closed before 31 January 2025. “After the cutoff” means the filing was published after it. That is a boundary reported by the provider, not an independent audit of the model's knowledge. Some readings were made after entry or after part of the return window; these are retrospective post-cutoff tests, not forecasts.

Spreads and ranks. The mean spread is the BUY-group average excess return minus the SELL-group average. The rank spread uses each reading's percentile within its fiscal year, so no single stock can dominate.

Permutation tests. Verdicts are shuffled 10,000 times within fiscal years, and separately within fiscal-year-and-industry groups (industry from a February 2026 FactSet snapshot; single-reading groups dropped). Two-sided p-values measure distance from the shuffled mean, with a plus-one correction; 0.0001 is the floor.

What was fixed in advance. The historical one-year and post-cutoff six-month tests were specified first. The post-cutoff one-year test, the rank statistic and the anachronism search were added after earlier results were seen. The verdict ladder, company grid and February 2026 options test follow `stories-config.yaml`, written before they were run. None is adjusted for multiple comparisons.

Ledgers. The 2025 scores are read from the origin project without modification; compounder dates come from file modification times, other dates from filenames. The February 2026 options scan uses each reading's stored timestamp.

Pipeline measurements. Page counts, characters and extraction times come from 40 randomly sampled cached 10-K PDFs; markup ratios and XBRL counts from 17 raw EDGAR 10-K files. Tokens are estimated at four characters each.

References

Sarkar and Vafa (2024) show that pretrained language models carry information about periods after their nominal analysis date, and test for it with events that should be unpredictable; the anachronism search in Section 9 is a simple instance. Glasserman and Lin (2023) measure look-ahead bias in model-scored news sentiment and reduce it by removing company names. Lopez-Lira and Tang (2023) find that model readings of headlines predict next-day returns; EON reads a different document over a longer horizon, and its results are not a replication.

Loughran and McDonald (2011) show why financial text needs domain-specific measurement: many words classed as negative by a general dictionary are neutral in financial disclosures. EON's richer readings still need the same comparison against simpler, explicit baselines.

- Loughran, T. and McDonald, B. (2011). *When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks*. Journal of Finance 66(1), 35–65. doi:10.1111/j.1540-6261.2010.01625.x.
- Google (2026). *Gemini API rate limits*. Provider documentation, accessed 29 September 2026.
- Glasserman, P. and Lin, C. (2023). *Assessing Look-Ahead Bias in Stock Return Predictions Generated by GPT Sentiment Analysis*. arXiv:2309.17322.
- Google (2026). *Gemini 2.5 Flash model documentation*, knowledge cutoff January 2025. Model documentation.
- Lopez-Lira, A. and Tang, Y. (2023). *Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models*. arXiv:2304.07619.
- Sarkar, S. K. and Vafa, K. (2024). *Lookahead Bias in Pretrained Language Models*. SSRN 4754678.
- US Securities and Exchange Commission (2026). *EDGAR fair access policy*: no more than ten requests per second.

Appendix B: Reproduction

Data availability. Evaluation tables, frozen prices, protocols, figure inputs and rendering scripts are retained in this repository's docs/whitepaper/ directory. The source databases and origin archive are local project inputs; this edition does not assert that they have been publicly released. SEC filings remain available through EDGAR. Appendix B maps the inputs needed to reproduce the results.

Table 3: Reproduction inputs and implementation entry points.

Path	Role
eon/data/sources/sec/	Download, Chrome conversion, extraction, SEC queue
eon/ai/	Prompts, key manager, request queue, Gemini provider
eon/ui/database/	Repository, migrations v001–v014
eon/ui/services/batch_queue.py	Leases, heartbeats, quota waits, alerts
evaluation-config.yaml	Main evaluation and its amendments
origin-ledger-config.yaml	2025 ledger test
stories-config.yaml	Ladder, company draw, February 2026 options test
sealed-ledger-config.yaml	The bet

From the EON root:

```
python docs/whitepaper/evaluate_backtest.py
python docs/whitepaper/evaluate_origin_ledger.py
python docs/whitepaper/evaluate_stories.py
.venv/bin/python docs/whitepaper/measure_pipeline.py
python docs/whitepaper/prepare_figure_data.py
python docs/whitepaper/render_figures.py
python docs/whitepaper/validate_paper.py
python docs/whitepaper/build_paper.py --pandoc /path/to/pandoc
cd docs/whitepaper && tectonic -X compile eon-whitepaper.tex
```

The evaluators open databases read-only and use saved prices; the origin evaluator reads `stock_stuff_06042025/10K_automator` without changing it. The sealed ledger's script refuses to run twice.

Prepared from the EON repository and its predecessors, their databases and files, and a frozen price snapshot. Negative and inconclusive results are retained. Research, not investment advice.

Figure source index

- Figure 1: data/eon.db; evaluation/trades.csv.
- Figure 2: the origin project; git history; data/archive/fintel.db; data/eon.db.
- Figure 3: the eon/ package.
- Figure 4: evaluation/pipeline-measurements.json; data/eon.db.
- Figure 5: eon/cli; eon/ui/services/batch_queue.py; eon/ai/request_queue.py.
- Figure 6: data/eon.db, frozen in figure-data/evidence.json.
- Figure 7: eon/ui/database/repository.py and migrations; eon/ui/services/batch_queue.py.
- Figure 8: the origin project's files; data/archive/fintel.db; data/eon_mac.db; data/eon.db.
- Figure 9: the evaluation protocols.
- Figure 10: evaluation/results.json.
- Figure 11: evaluation/stories.json.
- Figure 12: evaluation/trades.csv; evaluation/verification/bw-price-check.json.
- Figure 13: evaluation/stories.json.
- Figure 14: evaluation/origin-ledger/results.json.
- Figure 15: evaluation/stories.json.
- Figure 16: the origin files; evaluation outputs; data/eon.db; the flattening script's notes.
- Figure 17: evaluation/sealed-ledger/.